

SENTINEL-AI

Integrated CCTV & AI Traffic Management System

Gujarat Police Innovation Challenge 2026

The 80,000 Camera Problem

- 80K cameras generate ~320 Gbps of raw video data
 - Cost-prohibitive bandwidth requirements
- Fragmented systems across departments (Police, RTO)

Our Solution: Edge-AI Architecture

- Process video LOCALLY at the junction (NVIDIA Jetson)
- Transmit lightweight JSON metadata (96% bandwidth savings)
 - Unified GIS Command Center

Why We Stand Out (USPs)

- Edge-AI Bandwidth Eradication
- Zero-Latency eGujCop/VAHAN Database Correlation (<400ms)
 - Hardware Agnostic (Apple MPS + NVIDIA TensorRT)
 - Interactive Spatiotemporal Route Tracing

Key Feature: Live Video Wall

- Interactive WebSockets Dashboard
- Seamless Multi-Stream RTSP playback
- Expandable HUD with hardware telemetry

Key Feature: Route Reconstruction

- Input a suspect plate number (e.g. GJ01DX5432)
 - View chronological path on OpenStreetMap
 - Auto-generates court-admissible audit PDFs

Technical Stack

- AI: YOLOv11, NVIDIA TrafficCamNet, DeepStream
- Backend: Python, FastAPI, WebSockets, Kafka
- DB: PostgreSQL, Elasticsearch
- UI: HTML5, Leaflet.js

Thank You!

Ready for Deployment across Gujarat.

Demonstration ready on local feeds & Government RTSP Sandbox.